Chapter 9 - Practice Set

Conditional Statements

Instructions: Write each answer as a separate Python file and run it to verify your output. Attempt each problem on your own before reviewing the Chapter 9 notes.

Question 1

Write a program that asks the user to enter a number using input(). Use an if-elif-else chain to print a different message for each of the following cases: the number is positive, the number is negative, or the number is exactly zero. Also use a chained comparison to check whether the number falls between -10 and 10 (inclusive) and print an appropriate additional message if it does.

Question 2

Write a program that simulates a simple login check. Ask the user to enter a username and a password. Using nested if statements, check: (a) whether the username matches a stored username, and if so, (b) whether the password matches a stored password. Print a "Login successful" message only when both are correct, a "Wrong password" message when the username is correct but the password is wrong, and a "Username not found" message when the username itself does not match. Use at least two levels of nesting.

Question 3

Write a program that demonstrates the truthy and falsy behavior of the five empty containers you learned in Chapters 4-8. Create one empty variable for each type: an empty string, an empty list, an empty tuple, an empty set, and an empty dictionary. Use an if-else block for each one to print whether it is truthy or falsy without using len() or any comparison operator - rely solely on the variable itself as the condition (e.g., "if my_list:").

Question 4

Write a program that uses a dictionary to store three people's names as keys and their email addresses as values. Then prompt the user to enter a name. Use dict.get() to look up the email. Using "is None" and "is not None", check the return value and print either the email address that was found or a message saying the person is not in the directory. Do not use == None or != None anywhere in your solution.

Question 5

Write a program that demonstrates short-circuit evaluation. First, ask the user to enter a number. Using a single if statement with and, safely check whether the number is non-zero AND whether 100 divided by it is greater than 5, without causing a ZeroDivisionError when the user enters zero (rely on short-circuit evaluation to prevent the division). Then demonstrate the "or default" pattern: ask the user to enter their city, and if they press Enter without typing anything, assign and print "Unknown City" as the default using a single line with or.

Question 6

Write a program that asks the user to enter their score (0-100). Use a ternary expression (conditional expression) to assign "Pass" or "Fail" to a variable based on whether the score is 60 or above. Then use a second ternary expression to assign "Distinction" or "No Distinction" based on whether the score is 80 or above. Print both results. Do not use any multi-line if-else blocks for these two assignments - use only ternary expressions.

Question 7

Write a program that uses the match statement to build a simple calculator dispatcher. Ask the user to enter an operation name ("add", "subtract", "multiply", "divide"). Ask for two numbers. Use a match statement to perform the correct arithmetic operation based on the input and print the result. Include a case for "divide" that prints an error message if the second number is zero. Include a wildcard case _ that handles any unrecognized operation name.

Question 8

Write a program that uses the match statement with the | operator to classify the month a user enters (as a number 1-12) into one of four seasons: Spring (3, 4, 5), Summer (6, 7, 8), Autumn (9, 10, 11), and Winter (12, 1, 2). Each season should be a single case using | to match multiple month numbers. Include a wildcard case for any number outside 1-12 and print an appropriate error message.

Question 9

Write a program that simulates a ticket pricing system with nested conditions and multiple elif branches. Ask the user for their age and whether they are a student (yes or no). Apply the following rules: children under 5 are free; ages 5-17 pay a child price of $5; ages 18 and above pay the adult price of $15, UNLESS they are a student in which case they pay $10; ages 65 and above always pay a senior price of $8 regardless of student status. Print the ticket category and the price.

Question 10

Write a program that deliberately demonstrates three of the common beginner mistakes from Chapter 9, each caught and explained using a comment or a print statement. First, show a case where using separate if statements instead of elif produces multiple outputs for a single score (e.g., score = 85 printing both "B" and "C"). Then correct it using elif and show the difference. Second, show that "if my_list:" and "if len(my_list) != 0:" behave identically, but print a message explaining which is more Pythonic. Third, show a safe None check using both "== None" and "is None" and print a message explaining which style is preferred and why.

--- End of Chapter 9 Practice Set ---